Our Environment

OUR ENVIRONMENT

NCERT Class 10 Science • Ch 13 | Hinglish Detailed Notes

NCERT Class 10 Science – Chapter 13 (Hinglish Detailed Notes)

COLOUR CODE: RED = Core Basics / extra help (Class 7-8-9 wali cheezein jo yaad honi chahiye).

COLOUR CODE: GREEN = Board exam me ye topics BAAR-BAAR aate hain - pakka yaad karo.

Baaki normal black text = main chapter content (detail me, simple bhasha me).

CORE BASICS - PEHLE YE PADHO (CLASS 7-8-9 KA FOUNDATION)

Tune 7-8-9 nahi padhi - koi tension nahi. Ye page is chapter ki base hai.
Ek baar achhe se padh le, fir chapter aasaani se samajh aayega.

1) ENVIRONMENT (paryavaran) kya hai:
- Hamare aas-paas ka sab kuch = environment. Isme do cheezein hoti: LIVING (ped, jaanwar, insaan) + NON-LIVING (hawa, paani, mitti, dhoop).

2) ECOSYSTEM (paristhitik tantra) ka intro:
- Ek jagah ke saare jeev + non-living cheezein milke jo system banate, aur ek doosre se interact karte - usko ECOSYSTEM kehte (jaise talaab, jungle).

3) BIOTIC vs ABIOTIC (simple):
- BIOTIC = saare JEEVIT (living) - plants, animals, bacteria.
- ABIOTIC = NON-LIVING (nirjeev) - temperature, paani, mitti, light, hawa.

4) PRODUCER / CONSUMER / DECOMPOSER ka matlab:
- PRODUCER = khud apna khana banaye (green plants - photosynthesis se).
- CONSUMER = doosron ko kha ke energy le (jaanwar, insaan).
- DECOMPOSER = mre hue jeev/waste ko galaa ke khatam kare (bacteria, fungi).

5) FOOD CHAIN kya hai (Class 7-8 ka idea):
- "Kaun kisko khaata" ki ek line. Energy ek jeev se doosre me jaati. Jaise: ghaas -> hiran -> sher.

6) AUTOTROPH vs HETEROTROPH (recap):
- AUTOTROPH = khud bhojan banaye (plants). "Auto" = khud.
- HETEROTROPH = doosron par nirbhar (animals, fungi).

7) ENERGY kya hai - aur SUN sabka boss:
- Energy = kaam karne ki shakti. Saari food chains ki energy ka ULTIMATE (sabse mool) source = SOORAJ (Sun). Plant Sun se energy lete.

8) BIODEGRADABLE vs NON-BIODEGRADABLE (matlab):
- BIODEGRADABLE = jo bacteria/fungi se gal-sad jaaye (sabzi, kaagaz, gobar).
- NON-BIODEGRADABLE = jo na gale (plastic, kaanch, metal). Saalon tak rehta.

9) ATMOSPHERE aur OZONE (intro):
- ATMOSPHERE = Earth ke chaaro taraf gaiso ki chaadar (hawa ki parat).
- OZONE (O_3) = upar wali atmosphere me ek gas jo Sun ki harmful kirno (UV rays) se hamein bachati - jaise ek chhatri (umbrella).

1. ECOSYSTEM (PARISTHITIK TANTRA)

DEFINITION: Ek area ke saare jeev (biotic) aur unke aas-paas ki non-living cheezein (abiotic) milke, ek doosre se interact karte hue jo unit banate usko ECOSYSTEM kehte hain. (e.g. talaab, jungle, khet.)

1.1 ECOSYSTEM KE COMPONENTS (do hisse)

(A) BIOTIC COMPONENTS = saare JEEVIT (living):
- Producers (plants), Consumers (animals), Decomposers (bacteria/fungi).

(B) ABIOTIC COMPONENTS = saare NON-LIVING (nirjeev) factors:
- Temperature, paani (water), mitti (soil), light (dhoop), hawa (air).

Dono milke ecosystem chalaate - jeev abiotic par depend karte aur ulta bhi.

1.2 NATURAL vs ARTIFICIAL ECOSYSTEM

NATURAL (khud bana, kudrati) = insaan ne nahi banaya:
e.g. FOREST (jungle), POND (talaab), samundar, nadi.
ARTIFICIAL (manav-nirmit, man-made) = insaan ne banaya aur sambhalta:
e.g. AQUARIUM (machhli ghar), GARDEN (bagicha), CROP FIELD (khet).
Natural ecosystem khud-ba-khud balance rehta; artificial ko insaan ko maintain karna padta (paani badalna, khaad daalna, etc.).

2. ORGANISMS BY ROLE (KAUN KYA KAAM KARTA)

Ecosystem me har jeev ka ek "role" (kaam) hota - 3 main types:

2.1 PRODUCERS (UTPADAK)

- Green plants aur algae (kaai). Ye PHOTOSYNTHESIS se khud apna khana banate.
- Sun ki light + CO_2 + paani -> glucose (food) + O_2.
- Producers hi Sun ki energy ko food chain me LAATE hain (entry point).

2.2 CONSUMERS (UPBHOKTA)

- Jo khud khana nahi bana sakte, doosron ko khate. 4 type yaad rakho:
  - (i) HERBIVORES (shakahari) = sirf plants khate = PRIMARY consumer.
    e.g. hiran (deer), gaay (cow), bakri.
  - (ii) CARNIVORES (mansahari) = doosre jaanwar khate.
    - chhote/primary carnivore jo herbivore khaaye = SECONDARY consumer.
    - bade carnivore jo carnivore khaaye = TERTIARY consumer (sher, baaz).
  - (iii) OMNIVORES = plants aur animals DONO khaate (insaan, kauwa, bear).
  - (iv) PARASITES = doosre jeev (host) ke upar/andar rehkar usse food lete
    (e.g. juen, ticks, kuch keede). Host ko nuksaan hota.

2.3 DECOMPOSERS (APGHATAK) - bahut important

- BACTERIA aur FUNGI (kavak). Ye mre hue jeev / waste / patton ko GALA ke chhote-chhote simple padaarth me tod dete (decompose).

ROLE / IMPORTANCE of decomposers (exam me pakka):
- (i) Dead matter ko hata ke environment SAAF rakhte (warna laashein jam jaati).
- (ii) Mitti me NUTRIENTS (poshak tatva) wapas lautate -> RECYCLING karte.
- (iii) Inhi recycled nutrients ko plants dobara use karte. Cycle chalta rehta.

Agar decomposers na ho to: mra hua organic matter padaa rahega, mitti me nutrient wapas nahi aayenge, aur poora nutrient cycle ruk jaayega.

3. FOOD CHAIN aur FOOD WEB

FOOD CHAIN (aahar shrinkhala) DEFINITION: jeevo ki ek series jisme har jeev apne se pehle wale ko khaata - is tarah energy ek se doosre me transfer hoti.

EXAMPLE (yaad rakho): GHAAS -> HIRAN -> SHER
(grass -> deer -> lion). Teer (->) ka matlab "energy isko jaati / yeh usko khaata". Ghaas se energy hiran me, hiran se sher me.

3.1 TROPHIC LEVELS (POSHAK STAR)

Food chain ke har step ko TROPHIC LEVEL kehte. 4 level samjho:

- T1 = PRODUCERS (green plants) - first level, sabse niche.
- T2 = HERBIVORES / primary consumers (ghaas khane wale - hiran).
- T3 = chhote CARNIVORES / secondary consumers (hiran khane wale).
- T4 = bade CARNIVORES / tertiary consumers (top par - sher, baaz).

3.2 FOOD WEB (aahar jaal)

- Asli zindagi me ek jeev sirf ek hi cheez nahi khata. Bahut saari food chains aapas me jud jaati - inke is interconnected jaal ko FOOD WEB kehte.
  Food web = "bahut saari food chains ka net". Isse ecosystem stable rehta - agar ek khana khatam ho to jeev doosra option khaa lete.

4. FLOW OF ENERGY aur 10% LAW

- Energy food chain me Sun se aati: SUN -> PRODUCERS (plants) -> consumers.
- Energy ka flow UNIDIRECTIONAL hota = sirf EK hi direction me jaata (producer -> herbivore -> carnivore). Wapas niche nahi aata.

10% LAW (Lindeman ka niyam) - bahut important:

- Jab energy ek trophic level se AGLE level me jaati, to sirf lagbhag 10% energy hi transfer hoti hai. Baaki ~90% LOST ho jaati - heat ke roop me + saans/movement/life processes me kharch ho ke.

CHHOTA NUMERICAL idea:

Maano producers (T1) ke paas 1000 J energy hai, to -

T2 (herbivore) ko milegi = 100 J (10%)
T3 ko milegi = 10 J (10% of 100)
T4 ko milegi = 1 J (10% of 10)

FOOD CHAIN ME SIRF 3-4 LEVEL KYUN?

- Har step par 90% energy gum ho jaati. 4-5 level ke baad itni kam energy bachti ki ek aur jeev ko palna mushkil. Isliye chains chhoti (3-4 levels).

5. BIOLOGICAL MAGNIFICATION (BIOMAGNIFICATION)

DEFINITION: Kuch harmful chemicals (jaise PESTICIDES / keetnashak - DDT) jo gal nahi paate, wo food chain me upar jaane par concentration me BADHTE jaate hain. Isi ko BIOLOGICAL MAGNIFICATION kehte.

KAISE HOTA HAI:

- Khet me chhidke pesticides paani/mitti se plants me aate.
- Plant ko herbivore khaata - chemical uske body me jama.
- Us herbivore ko carnivore khaata - aur zyada jama. Aise har level par concentration badhti jaati (kyunki chemical body se nikalta nahi).

HUMANS SABSE ZYADA AFFECTED KYUN:

- Insaan aksar food chain ke TOP par hota (top consumer), isliye sabse zyada concentration insaan ke body me pahunchta - sabse zyada nuksaan.

Isiliye fields me zyada pesticide use karna lambe samay me khatarnaak hai.

6. OZONE LAYER aur uska DEPLETION

- OZONE (O3) = oxygen ke 3 atom wali gas. Ye atmosphere ki UPAR wali parat (stratosphere) me hoti hai.
- KAAM: Sun ki harmful ULTRAVIOLET (UV) kirno ko Earth tak aane se ROKTI - ek protective shield/chhatri ki tarah. Jeev-jantu ko UV se bachati.

OZONE DEPLETION (parat ka patla hona):

- Kuch man-made chemicals, khaaskar CFC (Chloro Fluoro Carbons), ozone ko tod dete. CFC aate hain - FRIDGE, AC (refrigerators), aur SPRAY/aerosol se.
- Inse ozone layer patli ho jaati (Antarctica ke upar "ozone hole" mila).

EFFECTS (UV zyada aane se nuksaan):

- Insaano me SKIN CANCER (twacha ka cancer), aankh ki bimari (cataract / EYE DAMAGE), immunity kamzor. Plants aur jal-jeevo ko bhi nuksaan.

MONTREAL PROTOCOL (1987):

- Duniya ke deshon ka agreement - CFC ka utpadan/use KAM karne ke liye, taaki ozone layer ki raksha ho.

7. WASTE aur uska MANAGEMENT

Hamare kachre (waste) ko 2 type me baanto:

7.1 BIODEGRADABLE WASTE

- Wo waste jo decomposers (bacteria/fungi) se gal-sad ke khatam ho jaata.
- Examples: bacha hua khana, sabzi-chilke, KAAGAZ (paper), COTTON (kapda), pedo ke patte, gobar (dung), lakdi.
- Problem: zyada matra me sade to badboo + harmful gases + ganndagi failti.

7.2 NON-BIODEGRADABLE WASTE

- Wo waste jo decomposers tod NAHI paate - saalon-saal padaa rehta.
- Examples: PLASTIC, kaanch (GLASS), METALS (dhaatu), DDT (pesticide).
- Problems: mitti/paani ki ganndagi (pollution), drains/nala block, jaanwar plastic kha ke marte, aur biomagnification (DDT jaise chemicals).

7.3 WASTE MANAGEMENT KE TARIKE (importance)

- SEGREGATION: kachre ko shuru me hi alag karo (biodegradable vs non-bio).
- RECYCLING: plastic, kaanch, metal ko dobara use/recycle karo.
- COMPOSTING: biodegradable waste se khaad (compost) banao - mitti ko faayda.

Sahi waste management se pollution kam hota, resources bachte aur environment saaf-suthra rehta. (Reduce, Reuse, Recycle yaad rakho.)

8. SOLVED EXAMPLES (HARDEST -> EASIEST)

Ye type ke questions exam me pakka aate - steps khud likhke practice karo.

EXAMPLE 1 (Hardest): 10% law numerical. Producers (T1) ke paas 20000 J energy.
Har aage wale trophic level (T2, T3, T4) ko kitni energy milegi?
Niyam: har level par sirf 10% energy aage jaati.
T2 (herbivore) = 10% of 20000 = 2000 J
T3 (sec. carniv) = 10% of 2000 = 200 J
T4 (ter. carniv) = 10% of 200 = 20 J
Answer: T2 = 2000 J, T3 = 200 J, T4 = 20 J.

EXAMPLE 2: Biomagnification kya hai aur insaan sabse zyada kyun affected hote?
Biomagnification = harmful chemicals (DDT/pesticides) ka food chain me upar jaane par concentration badhna (kyunki ye gal/nikal nahi paate). Insaan aksar food chain ke TOP par hai (top consumer), isliye sabse zyada chemical insaan ke body me jama hota -> sabse zyada nuksaan.

EXAMPLE 3: Food chain me sirf 3-4 trophic level kyun hote? (energy ke aadhar par)
Kyunki har trophic level par sirf ~10% energy hi aage jaati, ~90% heat aur life processes me lost. 4-5 step ke baad itni kam energy bachti ki aur jeev ko sustain karna sambhav nahi. Isliye chains chhoti rehti.

EXAMPLE 4: In items ko biodegradable / non-biodegradable me baanto aur 1 problem batao: (kaagaz, plastic bag, sabzi ke chilke, kaanch ki botal, gobar).
- Biodegradable: kaagaz, sabzi ke chilke, gobar (decomposers gala dete).
- Non-biodegradable: plastic bag, kaanch ki botal (gal-te nahi).

Problem: non-bio waste saalon padaa rehta -> pollution, drains block, jaanwar ko nuksaan. Bio waste zyada ho to badboo aur ganndagi.

EXAMPLE 5: Ozone depletion ka KAARAN aur ASAR (effect) batao.
Kaaran: CFC (Chloro Fluoro Carbons) jo fridge, AC, sprays se nikalte - ozone ko todte -> layer patli (ozone hole). Asar: zyada UV Earth par aata -> skin cancer, eye damage (cataract), immunity kamzor, fasal/jal-jeev ko nuksaan. (Bachav: Montreal Protocol se CFC kam karna.)

EXAMPLE 6: Decomposers ka role kya hai? Agar ye gayab ho jaayein to kya hoga?
Role: dead matter/waste ko galaa ke environment saaf rakhte aur nutrients mitti me wapas lautate (recycling). Agar gayab ho jaayein: mri hui cheezein jam jaati, nutrients recycle nahi hote, mitti banjar, aur poora nutrient cycle ruk jaata.

EXAMPLE 7 (Easiest): Is food chain me trophic levels pehchaano -
GHAAS -> TIDDA (grasshopper) -> MENDHAK (frog) -> SAANP (snake).
Answer: Ghaas = T1 (producer), Tidda = T2 (herbivore/primary consumer), Mendhak = T3 (secondary consumer), Saanp = T4 (tertiary consumer).

9. QUICK Q&A - PADHNE KE BAAD KHUD CHECK KARO

Pehle khud answer socho, fir niche milao. Ye board-style questions hain.

Q1. Ecosystem ke do main components kaunse?
A1. Biotic (saare living - plant, animal, decomposer) aur Abiotic (non-living - temperature, paani, mitti, light, hawa).

Q2. Producer, consumer aur decomposer me 1-1 line farak?
A2. Producer = khud khana banaye (green plant). Consumer = doosron ko khaaye (animal). Decomposer = dead matter ko galaa de (bacteria, fungi).

Q3. Decomposers ka ek bada importance batao.
A3. Dead matter ko todke nutrients ko mitti me wapas lautate (recycling) aur environment saaf rakhte.

Q4. Food chain aur food web me farak?
A4. Food chain = ek single line (ghaas->hiran->sher). Food web = bahut saari food chains ka aapas me juda hua jaal (network).

Q5. Trophic levels kya hain? T1 aur T2 kaun?
A5. Food chain ke steps. T1 = producers (plants), T2 = herbivores (primary consumers).

Q6. 10% law kya kehta hai?
A6. Ek trophic level se agle level me sirf ~10% energy transfer hoti, baaki ~90% heat aur life processes me lost ho jaati.

Q7. Energy ka flow unidirectional hai - matlab?
A7. Energy sirf ek hi direction me jaati (producer -> consumer -> top), wapas laut nahi sakti.

Q8. Biomagnification ek line me.
A8. Harmful chemicals (DDT) ka food chain me upar jaane par concentration badhna; top consumer (insaan) sabse zyada affected.

Q9. Ozone layer ka kaam aur depletion ka kaaran?
A9. Kaam: UV kirno se bachana. Kaaran: CFC (fridge/AC/spray se) ozone ko todte.

Q10. Biodegradable aur non-biodegradable ka 1-1 example.
A10. Biodegradable = kaagaz/sabzi-chilka/gobar. Non-biodegradable = plastic/kaanch/metal/DDT.

EXAM STRATEGY (1 minute revision):

- Ecosystem ke components (biotic + 5 abiotic) + natural vs artificial.
- Producer/consumer/decomposer + DECOMPOSER ka role (recycling) likhna aana.
- Food chain example + trophic levels (T1-T4) + food web ka matlab.
- 10% law + unidirectional flow + "3-4 level kyun" - ye pakka aata.
- Biomagnification (top consumer most affected) - short note ready rakho.

- Ozone: O_3 -> UV se bachav, CFC se depletion, Montreal Protocol.
- Biodegradable vs non-biodegradable + waste management (segregate/recycle/compost).
- CORE BASICS page (environment, biotic/abiotic, energy, Sun) bhool jaao to wapas pehle padho - base strong rakho.

Generated by Kiro for Krishna1k - All the best!